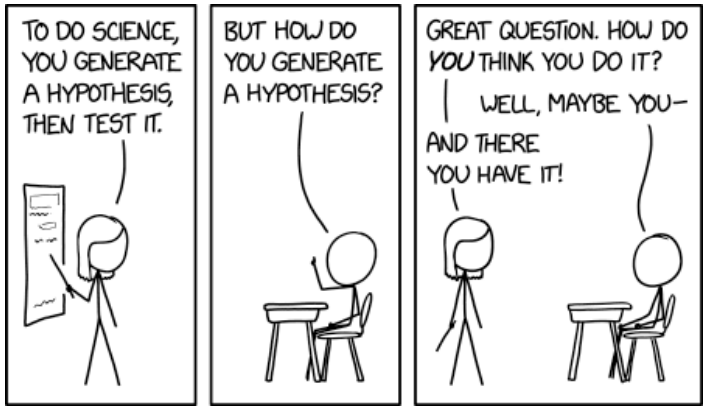
12 Hypothesis testing



Credit: explainxkcd

In **statistical testing** - decision making with uncertainty - we aim to use our collected data to test a *null hypothesis* about the population of interest. For example researchers screen genetic variants for associations with a phenotype, or gene expression levels for associations with disease.

Hypothesis testing quantifies how unusual the evidence is, **assuming that the null hypothesis is true.**

> **Important**
>
> Hypothesis testing compares the value of your statistic (sample mean, etc) to what we would expect to see if the null hypothesis were true. If the data are too unsual, compared to what we expect under the null, then the null hypothesis is rejected.

> **Note**
>
> **Example: is the coin fair?** Suppose we flip a coin 100 times to see if it is fair.

What is the null hypothesis?

What is the alternative hypothesis?

```
set.seed(0xdada)
numFlips = 100
probHead = 0.6
coinFlips = sample(c("H", "T"), size = numFlips,
  replace = TRUE, prob = c(probHead, 1 - probHead))
head(coinFlips)
```

```
[1] "T" "T" "H" "T" "H" "H"
```

```
table(coinFlips)
```

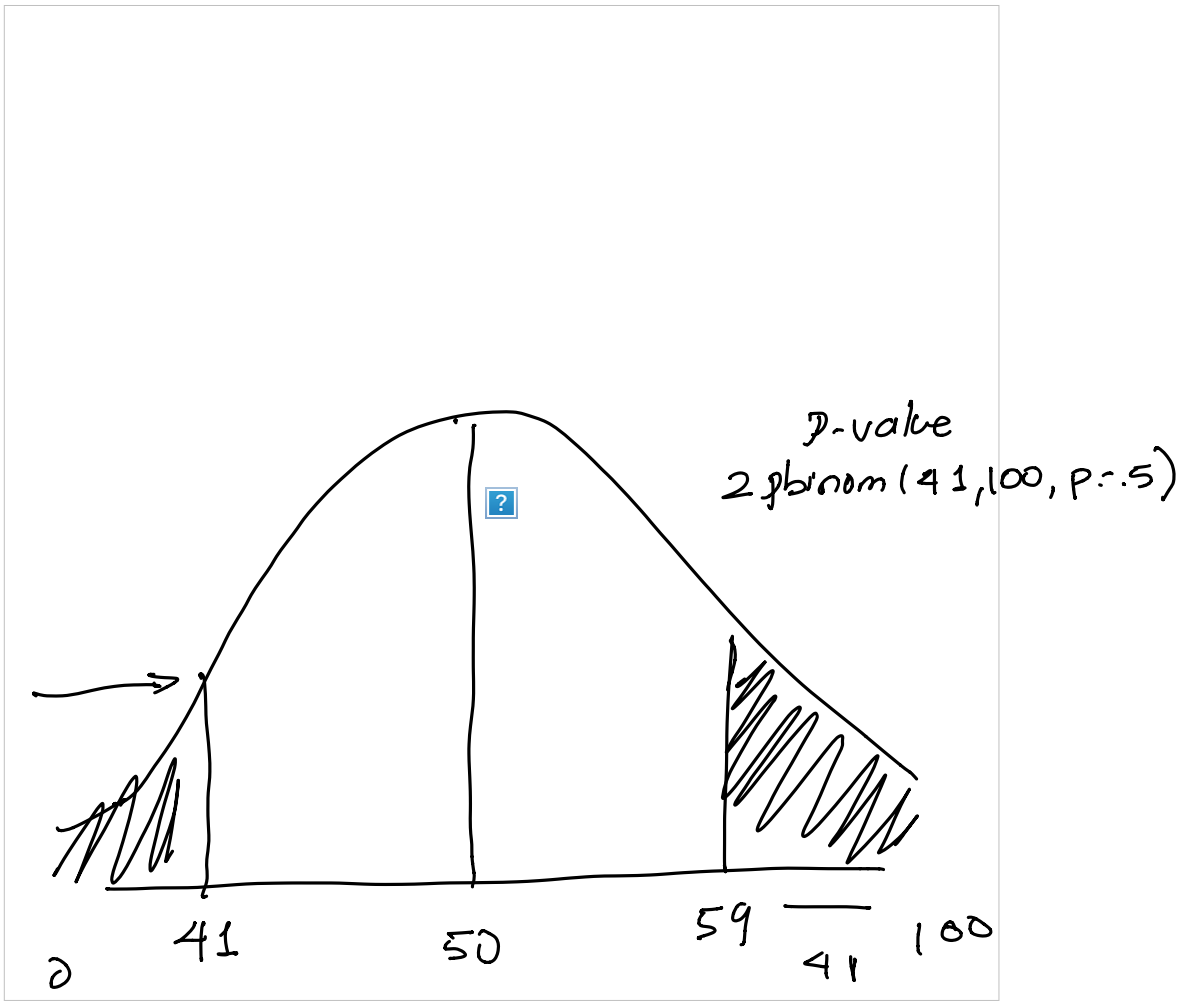
```
coinFlips
 H  T
59 41
```

$H_0: p = 0.5$

What is the null distribution for the number to heads? null distr. Binomial(100, p=.5)

```
library("dplyr")
k = 0:numFlips
numHeads = sum(coinFlips == "H")
binomDensity = tibble(k = k,
     p = dbinom(k, size = numFlips, prob = 0.5))
```

```
library("ggplot2")
ggplot(binomDensity) +
  geom_bar(aes(x = k, y = p), stat = "identity") +
  geom_vline(xintercept = numHeads, col = "blue")
```



How do we quantify whether the observed value, 59, is likely to be seen with a fair coin, or whether its deviation from the expected value (50) is already large enough for us to conclude with enough confidence that the coin is biased?

Choose a **significance level** α (e.g. 0.05). We define a rejection region R consisting of outcomes that are sufficiently unlikely under H_0, such that $\Pr_{H_0}(X \in R) \leq \alpha$. If the observed number of heads lies in R, we reject H_0;

otherwise we do not reject.

The significance level is the evidential threshold we agree upon at the outset.

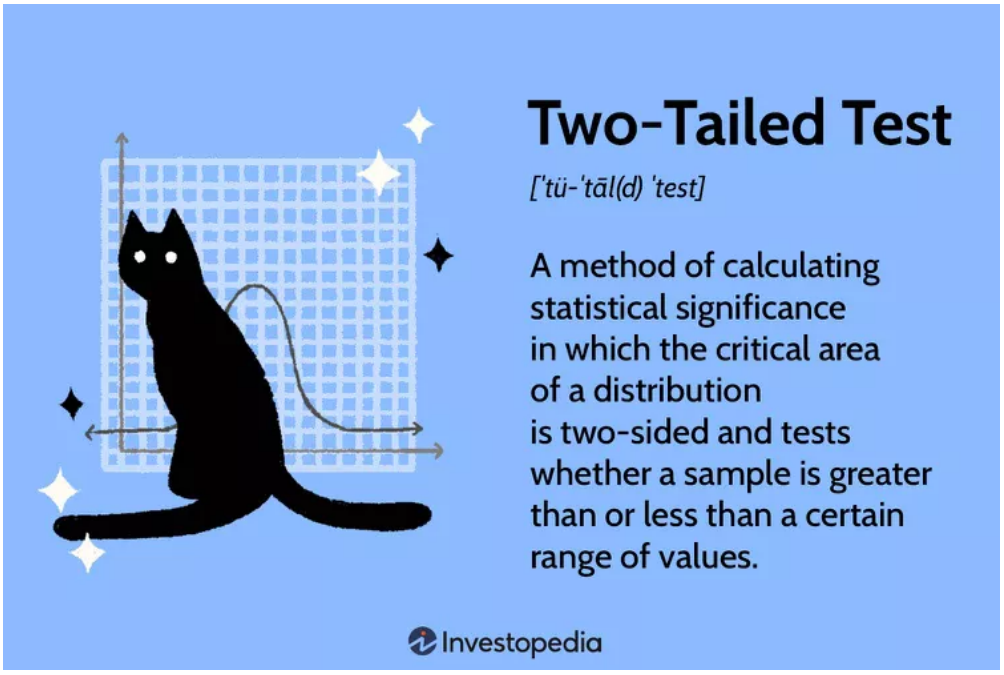
Sometimes the significance level is called the “alpha level” and denoted with α.

> **Important**
>
> **Reject or not to reject** Null and alternative hypotheses do not have equal standing. The null hypothesis is the one being tested with the data. If the data is inconsistent with the null hypothesis we reject, otherwise, we do not reject.

In our example, the alternative hypothesis includes parameter values on both sides of the parameter value specified by the null hypothesis. The test is referred as two-tailed.

```
library("dplyr")
alpha = 0.05
binomDensity = arrange(binomDensity, p) |>
        mutate(reject = (cumsum(p) <= alpha))

ggplot(binomDensity) +
  geom_bar(aes(x = k, y = p, col = reject), stat = "identity") +
  scale_colour_manual(
    values = c(`TRUE` = "red", `FALSE` = "darkgrey")) +
  geom_vline(xintercept = numHeads, col = "blue") +
  theme(legend.position = "none")
```



Credit: Investopedia

The **p-value** is the probability (assuming the null distribution) of observing a value (test statistic) at least as extreme as the one observed.

In our case, this value is:

```
2*pbinom(41,100,.5)
```

```
[1] 0.08862608
```

> **Important**
>
> **Punch line:** If the p-value of the data is less than or equal to α then the data are judged to provide enough statistically significant evidence in order reject the H_0 in favor of H_A.

Question 1. Does the fact that we don't reject the null hypothesis mean that the coin is fair?

Question 2. Would we have a better chance of detecting that the coin is not fair if we did more coin tosses? How many?

```
binom.test(x = numHeads, n = numFlips, p = 0.5)
```

```

	Exact binomial test

data:  numHeads and numFlips
number of successes = 59, number of trials = 100, p-value =
0.08863
alternative hypothesis: true probability of success is not
equal to 0.5
95 percent confidence interval:
 0.4871442 0.6873800
sample estimates:
probability of success
                  0.59
```

12.1 t-test

12.2 One population

You have data $x_1, \ldots, x_n$ realizations from a population mean μ. You want to test:

Null hypothesis: $H_0 : \mu = 0$ Alternative hypothesis (choose one):

Two-sided: $H_1 : \mu \neq 0$ Right-tailed: $H_1 : \mu > 0$ Left-tailed: $H_1 : \mu < 0$

Then the one-sample t statistic is

true value under the null

$$t = \frac{\bar{x} - 0}{s/\sqrt{n}} = \frac{\bar{x}}{s/\sqrt{n}}.$$

Under H_0 (and assuming approximate normality, or large n), this statistic follows

a t distribution with

$$\mathrm{df} = n - 1$$

degrees of freedom.

Compute t and the p-value from the t_{n-1} distribution depending on the alternative:

Two-sided: $p = 2P(T \geq \overset{t}{\cancel{|t|}})$ Right-tailed: $p = P(T \geq t)$ Left-tailed: $p = P(T \leq t)$

Reject H_0 if $p \leq \alpha$ (e.g., $\alpha = 0.05$).

Confidence interval connection:

A two-sided t-test at level α is equivalent to checking whether 0 lies in the $(1 - \alpha)$ CI:

$$\bar{x} \pm t_{1-\alpha/2,\, n-1} \frac{s}{\sqrt{n}}.$$

12.3 Two population difference

You have data $x_1, \ldots, x_n$ realizations from a population mean μ_x, and you have data $y_1, \ldots, y_n$ from a population with mean μ_y

Null hypothesis: $H_0 : \ \mu_1 - \mu_2 = 0$ Alternative hypothesis (choose one):

Two-sided: $H_1 : \ \mu_1 - \mu_2 \neq 0$ Right-tailed: $H_1 : \ \mu_1 - \mu_2 > 0$ Left-tailed: $H_1 : \ \mu_1 - \mu_2 < 0$

Then the two-sample t statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE_{(\bar{x}_1 - \bar{x}_2)}}$$

Under H_0 (and assuming approximate normality, or large n), this stat istic follows a t distribution with

$$\mathrm{df} = \mathrm{df} = \frac{\left(\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}\right)^2}{\frac{\left(\frac{s_x^2}{n_x}\right)^2}{n_x - 1} + \frac{\left(\frac{s_y^2}{n_y}\right)^2}{n_y - 1}}$$

degrees of freedom.

Compute t and the p-value from the t distribution depending on the alternative:

Two-sided: $p = 2P(T \geq |t|)$ Right-tailed: $p = P(T \geq t)$ Left-tailed: $p = P(T \leq t)$

Reject H_0 if $p \leq \alpha$ (e.g., $\alpha = 0.05$).

```
#stats::t.test(x, y, mu = 0)                  # Welch (default)
#stats::t.test(x, y, var.equal = TRUE)        # pooled equal-var
#stats::t.test(x, y, alternative="greater")   # H1: mu1 - mu2 > 0
```

What happens if you want to test: Null hypothesis: $H_0 : \ \mu_1 - \mu_2 = c?$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE_{(\bar{x}_1 - \bar{x}_2)}}$$

$H_a: \mu_1 - \mu_2 \neq c$

12.4 Randomization test (permutation t

$H_a: \mu_1 - \mu_2 < c$

This is a two-sample test that does not assume normality; it assumes the treatment labels were assigned at random (and units are independent).

You observe outcomes from n units, with n_1 labeled "treatment" and n_2 labeled "control". To test

$$H_0 : \ \text{no treatment effect}$$

Under H_0, the outcomes would be the same regardless of who was labeled treated, so the observed treatment labels are exchangeable. Therefore, any reassignment of the treatment labels (keeping n_1 treated) is equally likely.

Procedure

Choose a test statistic T measuring difference between groups (e.g., $\bar{Y}_T - \bar{Y}_C$) and compute the observed statistic T_{obs}.

Re-randomize: repeatedly shuffle the treatment labels (or enumerate all shuffles), recompute T each time. The p-value is the fraction of shuffles where T is at least as extreme as T_{obs} (two-sided uses $|T|$).

Example:

Outcomes: $y = [5, 6, 7, 8, 1, 2, 3, 4]$ Observed assignment: first 4 are treated, last 4 are control Treated: $[5, 6, 7, 8]$, Control: $[1, 2, 3, 4]$

Test statistic: difference in means

$$T = \bar{y}_T - \bar{y}_C.$$

Observed:

$$\bar{y}_T = 6.5, \quad \bar{y}_C = 2.5, \quad T_{\text{obs}} = 4.0.$$

Under H_0, any choice of 4 of the 8 units could have been labeled "treated" (there are $\binom{8}{4} = 70$ assignments). The randomization test:

computes T for each of the 70 possible labelings, compares them to 4.0.

```
y <- c(5,6,7,8,1,2,3,4)
z <- c(1,1,1,1,0,0,0,0)  # 1=treat, 0=control
```

```r
Tobs <- mean(y[z==1]) - mean(y[z==0])

set.seed(1)
B <- 10000
Tperm <- replicate(B, {
  zperm <- sample(z)  # shuffle labels (keeps 4 treated)
  mean(y[zperm==1]) - mean(y[zperm==0])
})

p_two_sided <- mean(abs(Tperm) >= abs(Tobs))
Tobs
```

```
[1] 4
```

```r
p_two_sided
```

```
[1] 0.025
```

Here, $T_{\text{obs}} = 4.0$ is actually the maximum possible difference in means (treated got the four largest outcomes), so the two-sided randomization p-value is

$$p = \frac{1}{70} \approx 0.0143$$

(only the observed assignment is as extreme

Reference: Modern Statistics for Modern Biology